



array

Level 2 · Array Search & Aggregate Operations

A set of Level 2 Verilog testbenches that operate on arrays — computing running sums, finding maximum and minimum values, and counting even/odd elements using for-loops and conditionals.

File type	Verilog testbench (.v)
Level	Level 2
Topics	Sum, max/min, even/odd count
Length	155 lines

```
module tb;
  integer array[9:0];
  integer i, sum;
  initial begin
    sum=0;
    for(i=0;i<10;i=i+1)begin
      array[i]=i+1;
      sum=sum+array[i];
    end
  end
endmodule
```

array — Level 2

```
/*=====
                        Array-based-problems
=====
*****
                        Level-2
*****

//Generate the array,it calculat the Sum
module tb;
integer array[9:0];
integer i;
integer sum;
initial begin
    sum=0;
    for(i=0;i<10;i=i+1)begin
        array[i]=i+1;
        sum=sum+array[i];
    $display("array[%0d]==%0d",i,sum);
    end
end
endmodule
OUTPUT:
# array[0]==1
# array[1]==3
# array[2]==6
# array[3]==10
# array[4]==15
# array[5]==21
# array[6]==28
# array[7]==36
# array[8]==45
# array[9]==55

//Find the maximum number, in give array
module tb;
integer array[9:0];
integer i;
integer max;
initial begin
    // Generate array
    for(i=0;i<10;i=i+1)begin
        array[i]=10*i;
    end
    //Assume frist element is max
    max=array[0];

    //Find the max
    for(i=0;i<10;i=i+1)begin
        if(array[i]>max)
            max=array[i];
    end

    //Print array elements
    $display("array[%0d]==%0d",i,array[i]);
    end
    $display("Maximum = %0d",max);
    end
endmodule
OUTPUT:
# array[0]=0
# array[1]=10
# array[2]=20
# array[3]=30
# array[4]=40
# array[5]=50
# array[6]=60
# array[7]=70
# array[8]=80
# array[9]=90
# Maximum = 90

//Find the minmum number, in give array
module tb;
integer array[9:0];
integer i;
integer min;
initial begin
    // Generate array
```

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for(i=0;i<10;i=i+1)begin
    array[i]=10*(i+1);
end
//Assume frist element is min
min=array[0];

//Find the min
for(i=0;i<10;i=i+1)begin
    if(array[i]<min)
        min=array[i];

//Print array elements
$display("array[%0d]=%0d",i,array[i]);
end
$display("Minmum = %0d",min);
end
endmodule

```

```

OUTPUT:
# array[0]=10
# array[1]=20
# array[2]=30
# array[3]=40
# array[4]=50
# array[5]=60
# array[6]=70
# array[7]=80
# array[8]=90
# array[9]=100
# Minmum = 10

```

```

//Count the even and odd numbers, give array
module tb;
integer array[9:0];
integer i;
integer even;
integer odd;
initial begin
    even=0;
    odd=0;
    //Genarate array
    for(i=0;i<10;i=i+1)begin
        array[i]=$urandom_range(10,50);
    end
    //find odd and even
    for(i=0;i<10;i=i+1)begin
        if(array[i]%2==0)begin
            even=even+1;
        end
        else begin
            odd=odd+1;
        end
        $display("array[%0d]=%0d",i,array[i]);
    end
    $display("Even=%0d",even); //count
    $display("odd=%0d",odd); //count
end
endmodule

```

```

OUTPUT:
# array[0]=29
# array[1]=18
# array[2]=34
# array[3]=26
# array[4]=32
# array[5]=13
# array[6]=19
# array[7]=30
# array[8]=43
# array[9]=46
# Even=6
# odd=4
*/

```